1. *Visual Data Analysis.* Given the dataset "task1-dataset.ods" (available in TeachCenter), which comprises a number of features. Provide a number of meaningful visualisations (4 visualisations) that show key properties of the dataset and dependencies. Based on the visualisations provide your interpretation and insights.

   (a) What pre-processing did your do? (e.g., Did you create new features? Did you normalise the data? Did you filter the dataset? Extended with another dataset?)

   (b) What are the most relevant dependencies between the features (selection of the figures)?

   (c) Provide a series of meaningful plots that show a specific relationship (dependency) or characteristic of the dataset

   (d) Provide a summary of the main insights

   Answer (a) - Preprocessing steps:

   - First preprocesing step
   - Second step
   - ...

   Answer (b) - List of main dependencies:

   - First dependency
   - ...

   Answer (b) and (c)

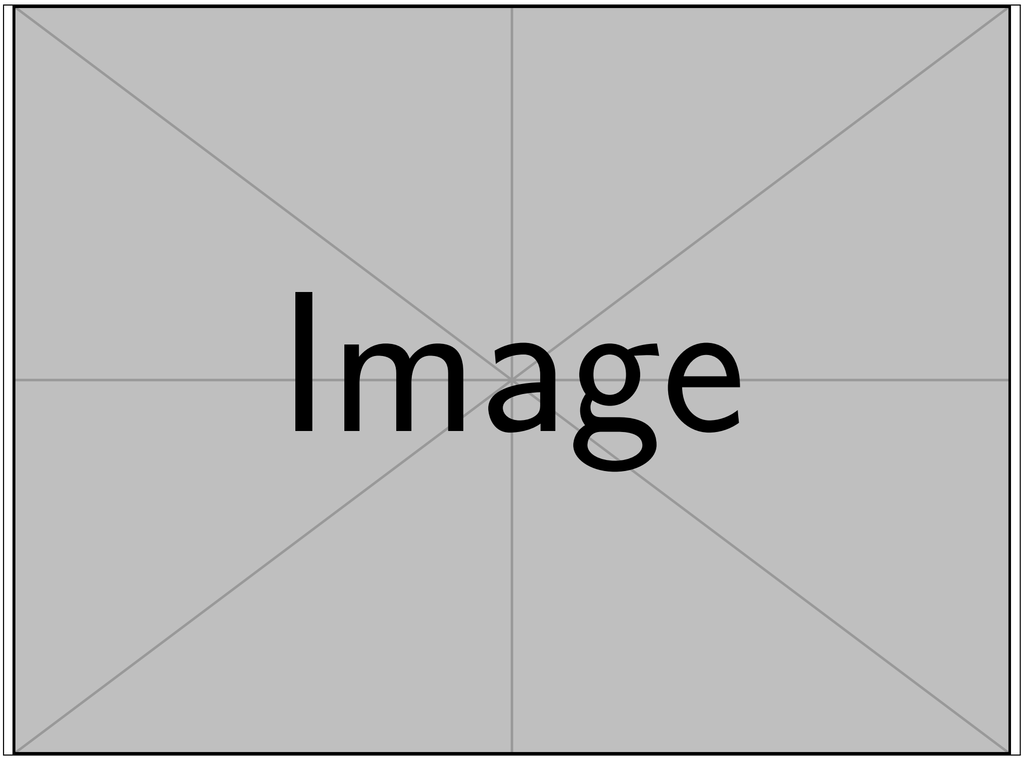


Figure 1: Please provide an explanation for the visualisation - (i) Why this kind of visualisation? (ii) What kind of dependency is being shown? (iii) What are potential interpretations?

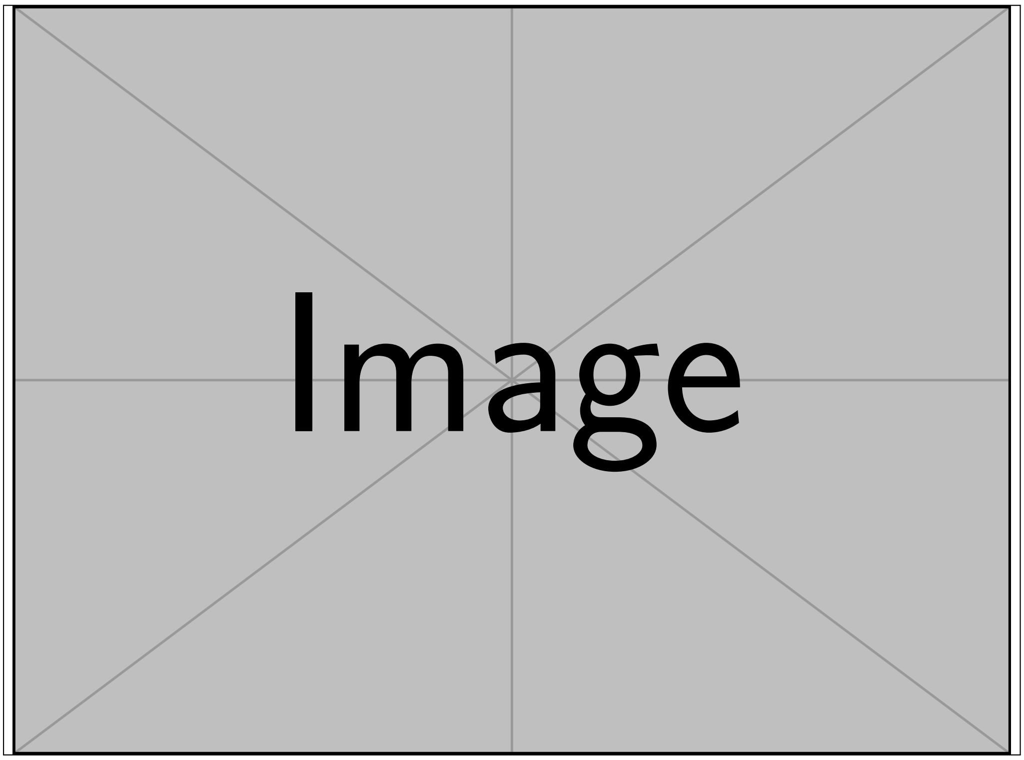


Figure 2: Please provide an explanation for the visualisation

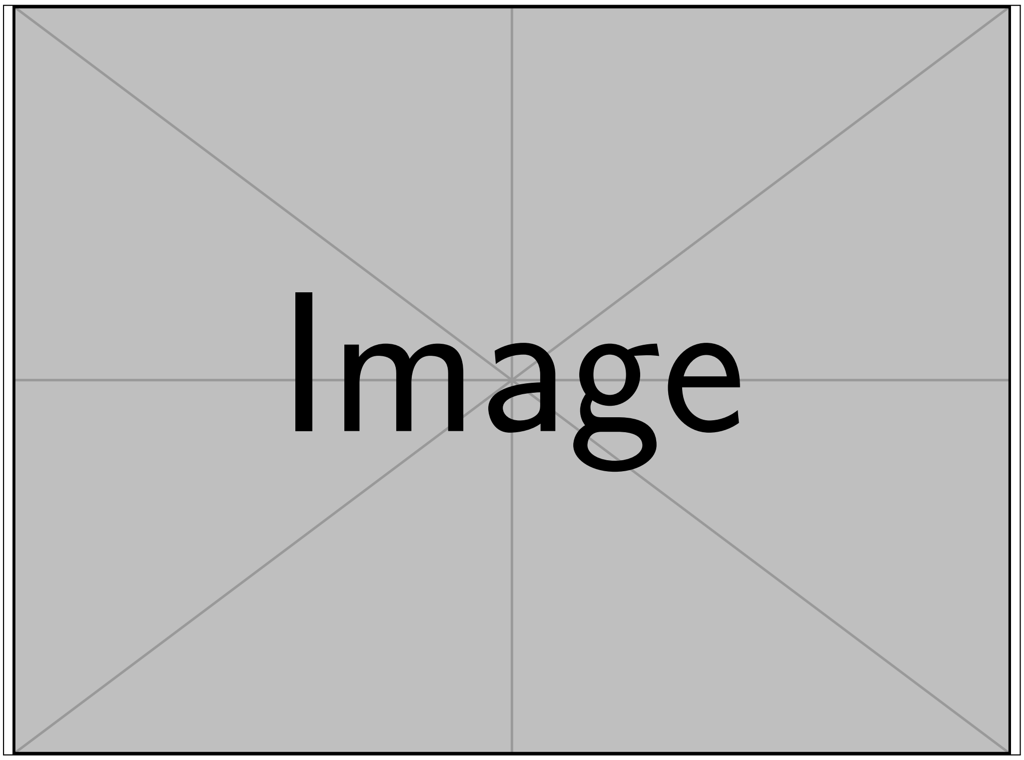


Figure 3: Please provide an explanation for the visualisation

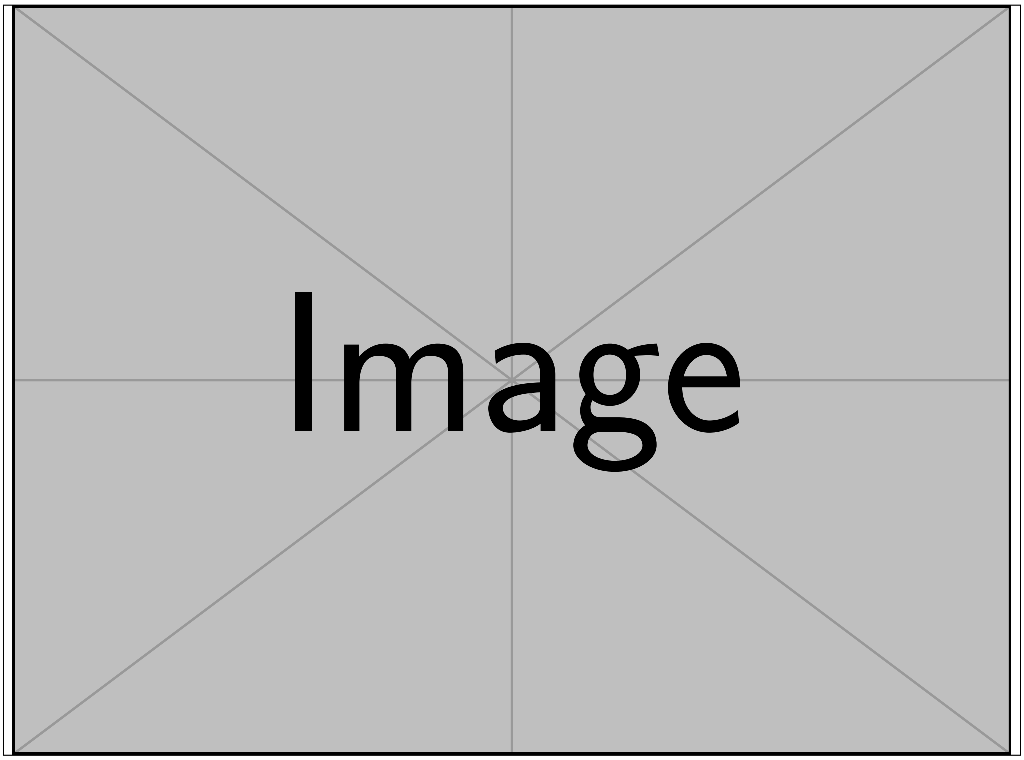


Figure 4: Please provide an explanation for the visualisation

Answer (d) - Short summary of the main insights (with references to the corresponding image)

- Finding #1, as illustrated in Figure 1 one ...
- ...

2. *Correlation.* Given a dataset, which consists of 1,000 variables (hint: most of them are just random), the goal is to find the relationships between variables, i.e., which and how do the variables relate to each other; what are the dependencies. The dataset "task2-dataset.csv" can be downloaded from TeachCenter.

   (a) Which methods did you apply to find the relationships, and why?

   (b) Which relationships did you find and how do you characterise the relationships (e.g., variable "Lurkowl (Strix umbra #1068)" to "Frosthawk (Accipiter glacies #1064)" is linear with correlation found via method X of 0.9)?

   (c) Which causal relationships between the variables can you find (e.g., variable "Rattlepuff (Lynx rattleus #1067)" causes "Slingshark (Carcharodon slingus #1068)")?

Answer (a) - Method and motivation:

(a) First Method - Why chosen (one sentence)

(b) Second Method

(c) ...

Answer (b)

Variable 1	**Variable 2**	**Type of dependency**	**Method**	**Value**
×	×	×	×	×
×	×	×	×	×

Answer (c)

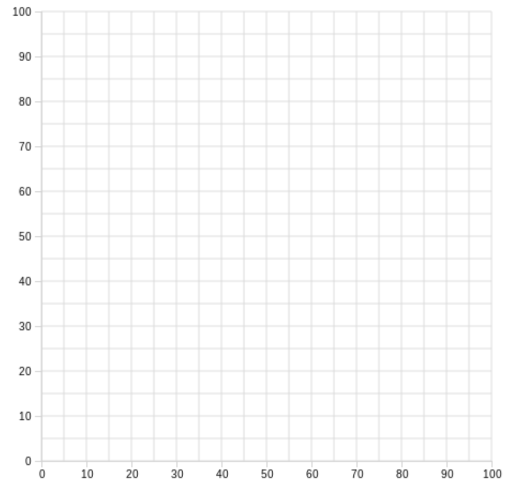
3. *Outliers/Anomalies.* Given two types of anomalies: (1) anomalies are defined to be **datapoints** in low density regions, (2) anomalies are **regions** of low density.

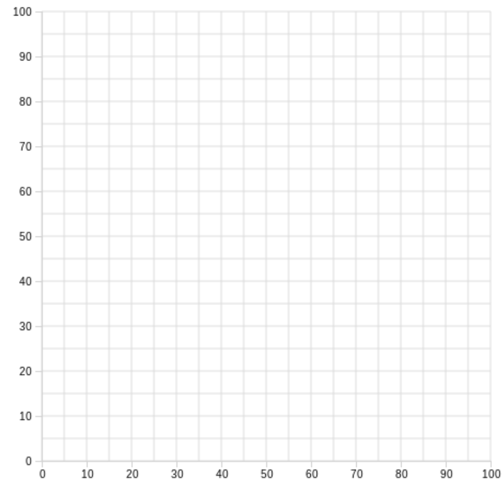
   (a) For both anomalies please create/draw a dataset of 2 features (x and y axis), with 3 anomalies and many normal data points (the normal datapoints should be marked, e.g., green colour)

   (b) Name the algorithms or describe the algorithmic way of how to identify this anomalous behaviour (you may also describe any necessary preprocessing)

   (c) Name the assumptions made by your algorithms

Answer (a) - Draw two datasets





Answer (b) - Describe the algorithms

Dataset 1 "Name of algorithm/method" followed by a justification, why the method has been selected...
Dataset 2 ...

Answer (c) - Describe the main assumptions

Algorithm	Assumption
×	×
×	×

4. *Missing Values.* The dataset "task4-dataset.csv" (available on TeachCenter) contains a number of missing values. Try to reconstruct why the missing values are missing? What could be an explanation?

   (a) What are the dependencies in the dataset?
   (b) What could be reasons for the missingness?
   (c) What strategies are applicable for the features to deal with the missing values?
   (d) For each feature provide an estimate of the arithmetic mean (before and after applying the strategies to deal with missing values)?

Answer (a) - Describe the dependencies in the dataset

X	Y	Type of dependency
×	×	×
×	×	×

Answer (b) - Describe the reason for missingness

Variable	Reason
×	×
×	×

Answer (c) - Describe the strategies for dealing with missing values

Variable	Strategy
×	×
×	×

Answer (d) - Arithmetic mean of original dataset (with the missing values), and the one after applying the strategies

Variable	Before Strategy	After Strategy
×	×	×
×	×	×